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Electrostatic input device

Abstract

An electrostatic input device includes a capacitance detector that is disposed at an operation part to be operated by a living body, and is configured to detect capacitance; and a controller that is formed of an integrated circuit, and is configured to transition the capacitance detector to an activated state in response to an increment in the capacitance detected by the capacitance detector reaching or exceeding a first predetermined value a predetermined number of times or more within a predetermined period in a case in which the capacitance detector is in a standby state and no contact with the living body is detected.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/JP2022/046332, filed on Dec. 16, 2022, and designated the U.S., which is based upon and claims priority to Japanese Patent Application No. 2022-000281, filed on Jan. 4, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND
1. Field of the Invention
(1) The present disclosure relates to electrostatic input devices.
2. Description of the Related Art
(2) Driving methods of input means using a capacitance detector provided at an operation part are known. According to the known driving methods, for saving power, the capacitance detector is set to be in a standby state in the absence of contact with the operation part, and the capacitance detector is set to be in an activated state in the presence of contact with the operation part (see, for example, US2004/0130527 A1).
SUMMARY
(3) An electrostatic input device according to an embodiment of the present disclosure includes: a capacitance detector that is disposed at an operation part to be operated by a living body, and is configured to detect capacitance; and a controller that is formed of an integrated circuit, and is configured to transition the capacitance detector to an activated state in response to an increment in the capacitance detected by the capacitance detector reaching or exceeding a first predetermined value a predetermined number of times or more within a predetermined period in a case in which the capacitance detector is in a standby state and no contact with the living body is detected.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating a door handle **10** attached to a vehicle **1**;
(2) FIG. **2** is a diagram illustrating an electrostatic sensor **110**;
(3) FIG. **3** is a diagram illustrating an electrostatic input device **100** of an embodiment;
(4) FIG. **4** is a diagram illustrating scanning intervals in a normal mode and a power saving mode;
(5) FIGS. **5**A and **5**B are diagrams illustrating outputs of sensors **111**, **112**, and **113**;
(6) FIG. **6** is a flowchart illustrating a process executed by a controller **126** in the power saving mode;
(7) FIG. **7** is a flowchart illustrating a process executed by the controller **126** in a first modified example of the embodiment;
(8) FIG. **8** is a diagram illustrating pulse waveforms used for scanning in the normal mode and the power saving mode in a second modified example of the embodiment;
(9) FIG. **9** is a flowchart illustrating a process executed by the controller **126** in the second modified example of the embodiment;
(10) FIG. **10** is a diagram illustrating pulse waveforms used for scanning in the normal mode and the power saving mode in a third modified example of the embodiment;
(11) FIG. **11** is a flowchart illustrating a process executed by the controller **126** in the third modified example of the embodiment;
(12) FIG. **12** is a diagram illustrating the outputs of the sensors **111**, **112**, and **113** in a fourth modified example of the embodiment; and
(13) FIG. **13** is a flowchart illustrating a process executed by the controller **126** in the fourth modified example of the embodiment.
DETAILED DESCRIPTION OF THE DISCLOSURE

(14) Existing driving methods of input means aim to reach a determination threshold in a short time that determines the event of contact. However, they do not necessarily achieve a speed that is high enough, and there has been scope for improvement. One possible measure against this is to lower the determination threshold that determines the event of contact with the operation part. In such a case, however, there is a possibility of increase in false detection as a result of activation due to, for example, instantaneous noise in response to motions of a user.
(15) In view of the above, the present disclosure provides an electrostatic input device that is capable of activating the capacitance detector in a state before determination of contact, and suppresses false detection.
(16) Hereinafter, embodiments in which the electrostatic input device of the present disclosure is applied will be described.
Embodiments
(17) FIG. **1** is a diagram illustrating the door handle **10** attached to the vehicle **1**. The door handle **10** is attached to a door **2** of the vehicle **1**. The door handle **10** has a thin and long shape so as to be readily grasped by fingers of a person's hand. The electrostatic sensor **110** is provided in the interior of the door handle **10**. In FIG. **1**, a longitudinal direction of the door handle **10** is a lateral direction in which the door handle **10** extends so as to be thin and long.
(18) FIG. **2** is a diagram illustrating the electrostatic sensor **110**. The electrostatic sensor **110** includes a substrate **110**A and the sensors **111**, **112**, and **113**. In one example, the sensors **111**, **112**, and **113** are arranged along the longitudinal direction of the door handle **10**. The electrostatic sensor **110** is an example of the capacitance detector.
(19) FIG. **3** is a diagram illustrating the electrostatic input device **100** of the embodiment. The electrostatic input device **100** includes the electrostatic sensor **110** and an integrated circuit **120**.
(20) The integrated circuit **120** includes a switch **121**, a switch **122**, an amplifier **123**, an ADC (Analog-to-digital converter: AD converter) **124**, an arithmetic unit **125**, and the controller **126**. The controller **126** includes a storage **126**A therein.
(21) As illustrated in FIG. **3**, the switch **121** of the integrated circuit **120** is connected to the sensors **111**, **112**, and **113** of the electrostatic sensor **110**.
(22) The switch **121** is provided between the sensors **111**, **112**, and **113** and an inverted input terminal of the amplifier **123**. When the controller **126** switches a connection destination, the inverted input terminal of the amplifier **123** is connected to the sensor **111**, **112**, or **113**. At a line branched from between the switch **121** and the inverted input terminal of the amplifier **123**, a power supply Vdd is connected via the switch **122**.
(23) The switch **122** is provided for applying a pulse voltage (voltage value: Vdd) to the sensors **111**, **112**, and **113** sequentially. This is performed by the controller **126** intermittently switching on/off of the switch **122** in a state in which the switch **121** is connected to the sensors **111**, **112**, and **113**.
(24) In this manner, when the controller **126** switches the connection destination of the switch **121** and switches on/off of the switch **122**, the pulse voltage is applied to the sensors **111**, **112**, and **113** sequentially. Applying the pulse voltage to the sensors **111**, **112**, and **113** is scanning the sensors **111**, **112**, and **113**.
(25) By scanning the sensors **111**, **112**, and **113** in this manner, the potentials detected by the sensors **111**, **112**, and **113** are amplified by the amplifier **123**, and converted by the ADC **124** from analog signals to digital signals. Based on the converted digital signals, the arithmetic unit **125** can calculate the capacitance between the sensors **111**, **112**, and **113** and a finger **200**. Information on the calculated capacitance is stored in the storage **126**A of the controller **126**. The controller **126** detects (determines) a user's approach to the door handle **10** in accordance with the information on the capacitance calculated by the arithmetic unit **125**.
(26) FIG. **4** is a diagram illustrating scanning intervals in the normal mode and the power saving mode. Here, the normal mode is a mode in which the electrostatic sensor **110** is activated at short

time intervals and detects the contact with or proximity to the living body. The electrostatic sensor **110** in the normal mode is referred to as being in the activated state.
(27) The power saving mode is a mode in which the electrostatic sensor **110** is in the standby state. In the standby state, the activation of the electrostatic sensor **110** is restricted. For example, by extending the scanning intervals or reducing the number of measurement pulses, the power consumed by the electrostatic sensor **110** and the integrated circuit **120** is set to be low. When a predetermined condition is fulfilled, the power saving mode is ended and transitioned to the normal mode, and the electrostatic sensor **110** is transitioned to the activated state from the standby state.
(28) In the normal mode, as an example, the sensors **111**, **112**, and **113** are scanned at 5 ms intervals, and a pulse voltage is sequentially applied to the sensors **111**, **112**, and **113** at 5 ms intervals. As illustrated in FIG. **4**, a single scanning time is a period for which the pulse voltage is applied to each of the sensors **111**, **112**, and **113** once.
(29) More specifically, the pulse voltage applied every 5 ms includes eight pulses as an example, as illustrated in the right-hand enlarged view. The timings of the pulse voltage applied to the sensors **111**, **112**, and **113** every 5 ms are temporally shifted from each other.
(30) The capacitance calculated by the arithmetic unit **125** in the single scanning process is an average value of eight capacitances obtained by application of eight pulses as illustrated in the enlarged view. That is, the number of times of measurement in the single scanning process is eight. Here, for simplifying the description, the number of times of measurement is set to be eight for the sake of convenience; however, in actual measurement, a larger number of times of measurement, e.g., 128 times, 256 times, or the like, may be set to obtain an average value. Conversely, a smaller number of times of measurement, e.g., less than eight times, such as once or the like, may be set.
(31) In the power saving mode, as an example, the sensors **111**, **112**, and **113** are scanned at 20 ms intervals, and a pulse voltage is sequentially applied to the sensors **111**, **112**, and **113** at 20 ms intervals. More specifically, the pulse voltage applied every 20 ms includes eight pulses as an example, as illustrated in the right-hand enlarged view. The timings of the pulse voltage applied to the sensors **111**, **112**, and **113** every 20 ms are temporally shifted from each other.
(32) Similar to the normal mode, a single scanning time is a period for which the pulse voltage is applied to each of the sensors **111**, **112**, and **113** once. The capacitance calculated by the arithmetic unit **125** in the single scanning process is an average value of eight capacitances obtained by application of eight pulses as illustrated in the enlarged view.
(33) In this manner, the number of times in which the pulse voltage is applied to the sensors **111**, **112**, and **113** is reduced in the power saving mode compared to the normal mode. Thus, the power consumed by the integrated circuit **120** is also reduced, and the power consumed by the electrostatic input device **100** can be reduced as a whole.
(34) In order to quickly activate the electrostatic sensor **110** from the power saving mode at an early stage before determination of approach, a possible way is to lower a threshold based on which the controller **126** detects (determines) the user's approach to the door handle **10** in accordance with the outputs of the sensors **111**, **112**, and **113**, and activate the electrostatic sensor **110** at the time of detection.
(35) However, in the state before the user's hand contacts the door handle **10**, the outputs of the sensors **111**, **112**, and **113** may contain noise due to change in the relative position between the user and the door handle **10**, disturbance noise, and the like. Even if filtering is performed by the controller **126**, it is challenging to accurately detect (determine) the user's approach to the door handle **10**.
(36) FIGS. **5**A and **5**B are diagrams illustrating the outputs of the sensors **111**, **112**, and **113**. FIG. **5**A is for comparison, illustrating a case of detecting (determining) the user's approach to the door handle **10** using a single threshold for the outputs of the sensors **111**, **112**, and **113**. Specifically, FIG. **5**A illustrates a relationship between the output of the sensor **111**, **112**, or **113** and the threshold in this case. FIG. **5**A is a characteristic diagram schematically illustrating the output of

the sensor **111**, **112**, or **113**. The average value of the outputs of the sensor **111**, **112**, or **113** shows similar characteristics.

(37) Here, the outputs of the sensors **111**, **112**, and **113** are signals to be input to the controller **126**. Specifically, signals representing potentials detected by the sensors **111**, **112**, and **113** are converted to a single signal as an output of the switch **121**, and amplified by the amplifier **123**. Subsequently, the signal is converted to a digital signal by the ADC **124**, and capacitance is obtained by the arithmetic unit **125**. That is, the outputs of the sensors **111**, **112**, and **113** are signals representing capacitance to be input to the controller **126**.

(38) As illustrated in FIG. **5**A, when the controller **126** detects (determines) a user' approach to the door handle **10** in accordance with the outputs of the sensors **111**, **112**, and **113** using a single low threshold set to detect the user's approach to the door handle **10**, the threshold is exceeded as a result of a single great change in the capacitance due to the influence of noise, unintended motions of the user, or the like. This may be detected as a user's approach to the door handle **10**, and can lead to false detection.

(39) Also, when the threshold is increased for avoiding false detection, there is a possibility that the electrostatic sensor **110** will be unable to be activated at a timing right enough for the user not to feel delay in reaction upon the approach of the user to the door handle **10**.

(40) In view of this, according to the electrostatic input device **100** of the embodiment, the electrostatic sensor **110** is activated in response to the increment in the capacitance detected by the electrostatic sensor **110** reaching or exceeding a predetermined value Ca (an example of the first predetermined value) the predetermined number of times or more within the predetermined period.

(41) FIG. **5**B is an example indicating a waveform obtained through scanning performed by the method as illustrated in FIG. **4** in the normal mode or the power saving mode, and illustrates fluctuations in the capacitances measured by the sensors **111**, **112**, and **113** in three scanning processes. The capacitance measured by the sensor **111**, **112**, or **113** in each of the scanning processes indicates an average value of the eight capacitances obtained by application of the eight pulses, as described above. The leftmost capacitance data in this graph is the most-recent scanning result. Here, for ease of understanding, description will be given with the capacitances measured by the sensors **111**, **112**, and **113** being simply treated as the same value, i.e., C**0**, assuming a state in which the user is not approaching the door handle **10** (non-operational state).

(42) In each of the scanning processes, three capacitances are measured by the three sensors **111**, **112**, and **113**. In the following description, capacitance increment ΔC**1**, ΔC**2**, or ΔC**3** obtained from the sensor **111**, **112**, or **113** in each of the first to third scanning processes is calculated by subtracting, from the maximum value of the capacitances measured in the current scanning process by the sensors **111**, **112**, and **113**, the capacitance measured in the previous scanning process by the sensor (**111**, **112**, or **113**) from which the maximum value of the capacitances was obtained.

(43) For the calculation of the increment, as an example, the maximum value of the increments in the capacitance for each of the sensors between the scanning processes may be used as a representative value of the increment between these scanning processes. Alternatively, the increment between the scanning processes may be calculated using an average or total value of the capacitances measured by the sensors in each of the scanning processes. Alternatively, the maximum value of the capacitances measured by the sensors in each of scanning processes may be treated as a representative value in that scanning process, and the increment between the scanning processes may be calculated using the representative value. Alternatively, an average or total value of the increments of the sensors between the scanning processes may be used as the increment between these scanning processes. A calculation method of the increments or the representative value of the capacitances in the scanning processes may be appropriately determined in accordance with a system employed.

(44) As an example, it is assumed that the capacitance measured by the sensor **111** is the maximum in the first scanning process, the capacitance measured by the sensor **112** is the maximum in the

second scanning process, and the capacitance measured by the sensor **113** is the maximum in the third scanning process. This assumes, for example, a case in which a person's hand, fingers, and the like approach the sensor **113** obliquely from above the sensor **111**.

(45) In this case, ΔC**1** is the value obtained by subtracting the capacitance C**0** of the most-recent scanning process (non-operational state) from the capacitance measured by the sensor **111** that is the maximum value in the first scanning process. Also, ΔC**2** is a value obtained by subtracting the capacitance measured by the sensor **112** in the first scanning process from the capacitance measured by the sensor **112** that is the maximum value in the second scanning process. Similarly, ΔC**3** is a value obtained by subtracting the capacitance measured by the sensor **112** in the second scanning process from the capacitance measured by the sensor **113** that is the maximum value in the third scanning process.

(46) As illustrated in FIG. **5**B, when the capacitance detected by the electrostatic sensor **110** increases three times or more within a period TP**1**, and ΔC**1**, ΔC**2**, and ΔC**3** in the three or more scanning processes for the increment in the capacitance are each a predetermined value Ca or more, the power saving mode is transitioned to the normal mode, thereby activating the electrostatic sensor **110**. At this time, ΔC**1**>Ca, ΔC**2**>Ca, and ΔC**3**>Ca.

(47) That is, in the example as illustrated in FIG. **5**B, the predetermined period is the period TP**1**, the predetermined number of times is three, and the capacitance increments used for the determination are ΔC**1**, ΔC**2**, and ΔC**3**. In order to simplify the description, an example in which the capacitance increments reach or exceed the predetermined value Ca in consecutive scanning processes will be described. However, a scanning process in which the increment does not exceed the predetermined value Ca or in which the capacitance decreases may be included between the scanning processes as described above. The scanning processes in which the increments reach or exceed the predetermined value Ca may be non-consecutive as long as the scanning processes in which the increments reach or exceed the predetermined value Ca within the predetermined period exist a predetermined number of times or more. In FIG. **5**B, the period TP**1** is illustrated as a period that is approximately equal to the period in which the three scanning processes are performed. However, the predetermined period may be set to be a longer period as long as the set period enables performing scanning processes in a larger number of times than the predetermined number of times. The example as illustrated in FIG. **5**B illustrates a case in which counting of the period TP**1** is started from the start point that is the first scanning process in which the capacitance increment exceeds Ca. However, after the period TP**1** is ended, counting of the period TP**1** may be started upon the capacitance increment exceeding Ca again. Also, as an example, at the time the capacitance increment exceeds Ca again before the end of counting of the period TP**1** started in the first scanning process (e.g., the second or third scanning process in FIG. **5**B), the following settings are possible: i.e., counting of the predetermined period may be started from that point in time serving as the start point, counting of the predetermined period may be performed in parallel with counting of the period TP**1** started in the first scanning process, and the number of times of proximity may be counted for each of the predetermined periods. At this time, when the number of times of proximity reaches or exceeds the predetermined number of times during counting of one of the predetermined periods, the power saving mode is transitioned to the normal mode. In addition, as an example, the start point of the period TP**1** may also be the time a parameter other than the capacitance increment exceeds a predetermined value. Also, the period TP**1** may be set to be counted at predetermined time intervals regardless of the predetermined value.

(48) The controller **126** includes a proximity event counter configured to count the number of proximity events. The proximity event counter is an example of the counter. When the capacitance detected by the electrostatic sensor **110** in the current scanning process increases by a value equal to or more than the predetermined value Ca compared to the capacitance detected in the previous scanning process, the controller **126** increments the proximity event counter. When the proximity event counter is three or more within a predetermined period (the period TP**1**), the controller **126**

activates the electrostatic sensor **110**.

(49) Hereinafter, fulfillment of the following condition will be referred to as fulfillment of an activation condition, i.e., in the power saving mode, the capacitance increments ΔC**1**, ΔC**2**, and ΔC**3** detected by the electrostatic sensor **110** reach or exceed the predetermined value Ca three times or more in the period TP**1**. When the activation condition is fulfilled, the controller **126** activates the electrostatic sensor **110**.

(50) According to the electrostatic input device **100** of the embodiment, when the electrostatic sensor **110** is in the standby state, the user's approach to the door handle **10** is detected (determined) using the determination method as illustrated in FIG. **5**B.

(51) In a case in which the activation condition is fulfilled and the electrostatic sensor **110** is activated, when the controller **126** determines that the output of the sensor **111**, **112**, or **113** in the normal mode exceeds the threshold that determines contact with the vehicle **1** (see FIG. **1**), the controller **126** determines contact with the door handle and transmits a signal permitting unlocking of a door lock to an ECU (Electronic Control Unit) configured to control the door lock of the vehicle **1**. As a result, the door lock is unlocked.

(52) FIG. **6** is a flowchart illustrating a process executed by the controller **126** in the power saving mode. The controller **126** repeatedly executes the flowchart as illustrated in FIG. **6** in a predetermined control cycle corresponding to the single scanning process as illustrated in FIG. **4**. Here, the capacitance calculated by the arithmetic unit **125** is converted from an analog value to a digital value and, for example, 10 pF becomes one count as a digital value.

(53) When the process is started, the controller **126** performs scanning of the sensors **111**, **112**, and **113** by controlling the connection of the switches **121** and **122** in the power saving mode (step S**1**).

(54) The controller **126** determines whether or not the capacitance calculated by the arithmetic unit **125** is equal to or more than a predetermined value Cwake that permits transition to the normal mode (step S**2**). The capacitance calculated at this time indicates the average value of the eight capacitances obtained by application of the eight pulses, as described above. Also, the predetermined value Cwake is set to be a value of capacitance large enough to accurately determine an increase in capacitance caused by the user's approach to the door handle **10** through single determination using a single threshold without causing false detection.

(55) When the controller **126** determines that the calculated capacitance is the predetermined value Cwake or more (S**2**: YES), transition to the normal mode is performed (step S**3**). Here, as an example, description will be given of a configuration in which a condition for transition to step S**3** is being equal to or more than the predetermined value Cwake representing the user's approach to the door handle **10**. However, when it can be reliably determined that the user has approached the sensors **111**, **112**, and **113**, such a condition may be used as the condition for transition to the normal mode.

(56) The controller **126** clears the proximity event counter stored in a memory to be 0 (step S**4**).

(57) When the controller **126** ends the process of step S**4**, the controller **126** ends the process in the control cycle corresponding to the current scanning timing (END).

(58) When the controller **126** determines in step S**2** that the calculated capacitance is not the predetermined value Cwake or more (S**2**: NO), the controller **126** determines whether or not the increment from the capacitance obtained in the previous scanning process is the predetermined value Ca or more (step S**5**).

(59) The determination in step S**5** is an operation corresponding to the example as illustrated in FIG. **5**B, i.e., a process that determines whether or not the maximum value of the three capacitances obtained by the sensors **111**, **112**, and **113** in the current scanning process has increased by the predetermined value Ca or more from the capacitance of the corresponding sensor (**111**, **112**, or **113**) in the previous scanning process.

(60) When the controller **126** determines that the increment has increased by the predetermined value Ca or more (S**5**: YES), the controller **126** increments the proximity event counter, i.e., adds

one to the count value stored in the memory, followed by overwriting (step S**6**).
(61) The controller **126** determines whether or not the proximity event counter is three or more (step S**7**).
(62) When the controller **126** determines that the proximity event counter is three or more (S**7**: YES), the controller **126** causes the flow to proceed to step S**3** for performing transition to the normal mode.
(63) When the controller **126** determines in step S**5** that the increment has not increased by the predetermined value Ca or more (S**5**: NO), the controller **126** determines whether or not the time from the start of counting, i.e., the duration time, is equal to or longer than the period TP**1** (step S**8**).
(64) When the controller **126** determines that the duration time is equal to or longer than the period TP**1** (S**8**: YES), the controller **126** clears the proximity event counter stored in the memory to be 0 (step S**9**).
(65) When the controller **126** determines in step S**7** that the proximity event counter is not three or more (S**7**: NO), the controller **126** ends the process in the current control cycle (END). Then, the controller **126** repeats the process from step S**1** in a control cycle corresponding to the next scanning timing.
(66) When the controller **126** determines in step S**8** that the duration time is not equal to or longer than the period TP**1** (S**8**: NO), the controller **126** ends the process in the control cycle corresponding to the current scanning timing (END).
(67) In the process as described above, transition to the normal mode is performed when the capacitance detected by the electrostatic sensor **110** increases as a result of the approach of the user to the door handle **10** in the power saving mode, and the increment in capacitance from the capacitance in the previous scanning process reaches or exceeds the predetermined value Ca three times or more within the period TP**1**.
(68) Therefore, it is possible to suppress false detection due to noise, unintended motions of the user, or the like, and quickly transition the power saving mode to the normal mode as the user's hand, fingers, and the like approach the electrostatic sensor.
(69) Accordingly, it is possible to provide the electrostatic sensor **110** that is quickly activated from the standby state and is suppressed in false detection.
(70) In the above, a configuration in which the electrostatic sensor **110** includes the three sensors **111**, **112**, and **113** has been described. The number of sensors may be two or may be four or more. In these cases, a control process performed by the controller **126** is the same as that in which the electrostatic sensor **110** includes the three sensors **111**, **112**, and **113**.
(71) The electrostatic sensor **110** may include a single sensor (e.g., the sensor **111**). In this case, when the capacitance obtained by the sensor **111** reaches or exceeds the predetermined value Ca three times or more within the period TP**1** compared to the capacitance in the previous scanning process, the power saving mode may be transitioned to the normal mode.
(72) In step S**5**, the proximity event counter is incremented when the value of the capacitance increases by the predetermined value Ca or more from the time of the previous scanning process. However, the proximity event counter may be incremented when the capacitance increment increases over time. Specifically, as illustrated in FIG. **5**B, the capacitance increment obtained in the first scanning process within the period TP**1** is denoted by ΔC**1**, the capacitance increment obtained in the second scanning process within the period TP**1** is denoted by ΔC**2**, and the capacitance increment obtained in the third scanning process within the period TP**1** is denoted by ΔC**3**. Here, in step S**5**, when ΔC**1** is the predetermined value Ca or more in the first scanning process, a relation of $\Delta C1<\Delta C2$ is fulfilled in the second scanning process, or a relation of $\Delta C2<\Delta C3$ is fulfilled in the third scanning process, the proximity event counter may be incremented. In this case, a scanning process in which the increment does not exceed the predetermined value Ca or the capacitance decreases during the period TP**1** may be included. A

possible method for confirming this is as follows. Specifically, the capacitance values in the scanning processes are stored in the memory. The increments in the scanning processes that are the predetermined value Ca or more are denoted by ΔC**1**, ΔC**2**, and ΔC**3**. Fulfillment of a relation of ΔC**1**<ΔC**2**<ΔC**3** within the period TP**1** is determined. When the relation is fulfilled, transition to the normal mode is performed. The same applies to subsequent ΔC**1**, ΔC**2**, and ΔC**3**.
(73) In the above description, a case in which the increment increases in the first, second, and third scanning processes has been described. However, the predetermined value for comparison may be increased gradually and compared. Specifically, when Ca<ΔC**1**, Cb<ΔC**2**, Cc<ΔC**3**, and Ca<Cb<Cc are fulfilled, transition to the normal mode may be performed.
(74) Also, a threshold may be provided for the difference between the capacitance increments obtained within the period TP**1**. Specifically, as illustrated in FIG. **5**B, when the capacitance increments ΔC**1**, ΔC**2**, and ΔC**3** obtained three times within the period TP**1** each increase over time by the predetermined value Ca or more, transition to the normal mode may be performed. This means fulfillment of ΔC**2**−ΔC**1**>Ca and ΔC**3**−ΔC**2**>Ca. This case indicates that the hand is approaching with an increasing speed. When the hand is approaching with an increasing speed, transition to the normal mode may be performed earlier. That is, when the hand is approaching with an increasing speed, a low threshold may be used such that transition to the normal mode is performed even if the hand is not close. This determination is performed in step S**5**. That is, when the controller **126** determines that a relation of ΔCn+1−Δn>Ca (where n is a natural number) is fulfilled (S**5**: YES), the controller **126** increments the proximity event counter, i.e., adds one to the count value stored in the memory, followed by overwriting (step S**6**). Meanwhile, when the controller **126** determines in step S**5** that a relation of ΔCn+1−Δn>Ca is not fulfilled (S**5**: NO), the controller **126** determines whether or not the time from the start of counting, i.e., the duration time, is equal to or longer than the period TP**1** (step S**8**).
(75) Also, when the difference between the capacitance increments is Ca or more and increases over time such that a relation of ΔC**3**−ΔC**2**>ΔC**2**−ΔC**1**>Ca is fulfilled, transition to the normal mode may be performed.
(76) In other words, when the difference between the capacitance increments increases over time and increases stepwise from Ca to Cb (>Ca) such that relations of ΔC**2**−ΔC**1**>Ca, ΔC**3**−ΔC**2**>Cb, and Cb>Ca are fulfilled, transition to the normal mode may be performed.
(77) Regarding step S**8**, a configuration in which the proximity event counter is cleared after the period TP**1** has passed, has been described. However, for example, when a very large increment is obtained, such as when an increment from the capacitance obtained in the previous scanning process is two or more times larger than the predetermined value Ca, even if the value obtained in the current scanning process is less than the predetermined value and the period TP**1** has passed, the count value may not be decremented or changed to one without clearing the proximity event counter. This is because of enabling quicker transition to the normal mode (activating the electrostatic sensor **110**) at the time of the next determination in view of the individual differences in the way to approach the vehicle **1**. The value that is two times larger than the predetermined value Ca is an example of a third predetermined value. The decremented count value "1" or the unchanged count value is an example of a second predetermined count value.
(78) FIG. **7** is a flowchart illustrating a process executed by the controller **126** in the first modified example of the embodiment. The flowchart as illustrated in FIG. **7** is the same as the flowchart as illustrated in FIG. **6** except for addition of a transition process from the normal mode to the power saving mode, i.e., a process of steps S**11** to S**15**. Thus, only the difference from FIG. **6** will be described.
(79) When the process is started, the controller **126** determines whether or not the power saving mode is set (step S**11**).
(80) When the controller **126** determines that the power saving mode is not set (step S**11**: NO), the controller **126** performs a scanning process in the normal mode (step S**12**). That is, as illustrated in

FIG. **4**, scanning is performed at 5 ms cycles.
(81) The controller **126** determines whether or not the condition for transition to the power saving mode is fulfilled (step S**13**). The condition for transition to the power saving mode is that the capacitance obtained by the sensors **111**, **112**, and **113** over the past 30 seconds is equal to or less than a predetermined value indicating no contact with the door handle **10**. The 30 seconds is an example of a predetermined time included in the condition for transition to the power saving mode.
(82) When the controller **126** determines that the condition for transition to the power saving mode is fulfilled (S**13**: YES), the controller **126** performs transition to the power saving mode (step S**14**).
(83) The controller **126** clears the proximity event counter (step S**15**). When the controller **126** ends the process of step S**15**, the controller **126** ends the flow (END).
(84) When the controller **126** determines in step S**13** that the condition for transition to the power saving mode is not fulfilled (S**13**: NO), the controller **126** ends the flow (END).
(85) When the controller **126** determines in step S**11** that the power saving mode is set (S**11**: YES), the controller **126** causes the flow to proceed to step S**1**. Subsequently, the process proceeds in the same manner as in the flowchart as illustrated in FIG. **6**.
(86) In a manner in which setting to the power saving mode is first determined as illustrated in FIG. **7**, when the controller **126** determines that the power saving mode is not set (S**11**: NO), scanning in the normal mode is performed.
(87) FIG. **8** is a diagram illustrating pulse waveforms used for scanning in the normal mode and the power saving mode in the second modified example of the embodiment.
(88) In the normal mode, similar to FIG. **4**, the sensors **111**, **112**, and **113** are scanned at 5 ms intervals as an example. The pulse voltage applied every 5 ms includes eight pulses as an example, as illustrated in the right-hand enlarged view. The capacitance calculated by the arithmetic unit **125** in a single scanning process is an average value of the eight capacitances obtained by application of the eight pulses as illustrated in the enlarged view.
(89) Meanwhile, in the power saving mode, the sensors **111**, **112**, and **113** are scanned at 20 ms intervals as an example. However, the pulse voltage applied every 20 ms includes four pulses as an example, as illustrated in the right-hand enlarged view. The capacitance calculated by the arithmetic unit **125** in a single scanning process is an average value of the four capacitances obtained by application of the four pulses as illustrated in the enlarged view. That is, the number of times of measurement in the single scanning process is four.
(90) In this manner, by reducing the number of times of measurement in the single scanning process, the power consumed in the power saving mode can be further reduced.
(91) FIG. **9** is a flowchart illustrating a process executed by the controller **126** in the second modified example of the embodiment. The flowchart as illustrated in FIG. **9** is the same as the flowchart as illustrated in FIG. **7** except that the processes in steps S**3** and S**14** are changed. Thus, only the difference from FIG. **7** will be described.
(92) When the controller **126** determines that the condition for transition to the power saving mode is fulfilled (S**13**: YES), the controller **126** performs transition to the power saving mode (step S**14**A). In step S**14**A, as illustrated in the enlarged view of the power saving mode in FIG. **8**, the number of times of measurement in the single scanning process is reduced to four.
(93) When the controller **126** determines in step S**2** that the capacitance is the predetermined value Cwake or more (S**2**: YES), the controller **126** performs transition to the normal mode (step S**3**A). In step S**3**A, the number of times of measurement in the single scanning process is changed to eight in the normal mode.
(94) As described above, in the power saving mode, the number of times of measurement in the single scanning process may be reduced compared to that in the normal mode. Although the manner in which the number of times of measurement is reduced to four has been described here, the number of times of measurement may be as desired, as long as it is less than the number of times of measurement in the normal mode.

(95) FIG. **10** is a diagram illustrating pulse waveforms used for scanning in the normal mode and the power saving mode in the third modified example of the embodiment.
(96) In the power saving mode, the capacitance is measured in a state in which the user is approaching the door handle **10**. Thus, the measurement can be influenced by disturbance noise and the like. In such a case, the frequency at which the pulses used for performing measurement multiple times in the single scanning process are output may be changed to a frequency that is less influenced by noise. Here, what is called frequency hopping is utilized for changing the frequency.
(97) In order to select a frequency that is less influenced by noise, it is necessary to detect such a frequency that is less influenced by noise. The level of noise can be estimated by continuously measuring capacitances of the same frequency in a very short cycle and observing the difference between them. According to the third modified example, as illustrated in the enlarged view of FIG. **10**, in the normal mode, solid-line eight pulses for measurement are output, followed by changing the frequency and outputting dashed-line six pulses for detecting the frequency that is less influenced by noise. Here, measuring the capacitance with the dashed-line pulses is referred to as noise scanning.
(98) When such noise scanning is performed, the frequency is less influenced by noise as the capacitance calculated by the arithmetic unit **125** is lower. The dashed-line six pulses are those of three different frequencies, and two pulses of the same frequency are output consecutively.
(99) The measurement is performed with six pulses for noise scanning, and the frequency of the pulse at which the capacitance is the lowest may be set to the frequency of the eight pulses for the measurement in the next scanning.
(100) When utilizing such frequency hopping, in the power saving mode, solid-line eight pulses for measurement are output, followed by changing the frequency, and dashed-line four pulses for noise scanning are output. That is, the number of the dashed-line pulses for noise scanning is smaller by two than in the normal mode.
(101) Thus, in the power saving mode, the power consumption can be reduced by reducing the number of pulses for noise scanning. The number of values of the frequency and the number of pulses as described in FIG. **10** are examples, and can be appropriately changed.
(102) FIG. **11** is a flowchart illustrating a process executed by the controller **126** in the third modified example of the embodiment. The flowchart as illustrated in FIG. **11** is the same as the flowchart as illustrated in FIG. **9** except that step S**21** is added and the processes of steps S**3** and S**14** are changed. Thus, only the difference from FIG. **9** will be described.
(103) When the process is started, the controller **126** sets the frequency of pulses for measurement from the result of noise scanning (step S**21**). When the process of step S**21** is performed for the first time, the result of noise scanning does not exist, and thus the frequency of pulse for noise scanning may be set to a predetermined initial value. When the controller **126** ends the process of step S**21**, the controller **126** causes the flow to proceed to step S**11**.
(104) In step S**14**B, the controller **126** reduces the number of pulses for noise scanning to four upon transition to the power saving mode (step S**14**B).
(105) In step S**3**B, the controller **126** increases the number of pulses for noise scanning to six upon transition to the normal mode (step S**3**B).
(106) As described above, when performing noise scanning utilizing frequency hopping, the power consumption may be reduced by reducing the number of pulses for noise scanning in the power saving mode.
(107) FIG. **12** is a diagram illustrating outputs of the sensors **111**, **112**, and **113** in the fourth modified example of the embodiment. In the fourth modified example, as illustrated in FIG. **5**B, when the capacitance increments ΔC**1**, ΔC**2**, and ΔC**3** over three times or more increase by the predetermined value Ca or more over time, the cumulative value of the differences between the increments is used to determine whether or not to perform transition, instead of performing transition to the normal mode.

(108) As illustrated in FIG. **12**, when the sensors **111**, **112**, and **113** are scanned four times, it is assumed that the capacitance increments obtained by the sensors **111**, **112**, and **113** in the first scanning process are ΔC**11**, ΔC**12**, and ΔC**13**. Similarly, it is assumed that the capacitance increments obtained by the sensors **111**, **112**, and **113** in the second scanning process are ΔC**21**, ΔC**22**, and ΔC**23**, and the capacitance increments obtained by the sensors **111**, **112**, and **113** in the third scanning process are ΔC**31**, ΔC**32**, and ΔC**33**. It is also assumed that the capacitance increments obtained by the sensors **111**, **112**, and **113** in the fourth scanning process are ΔC**41**, ΔC**42**, and ΔC**43**.
(109) In this case, when the cumulative value of the differences between: the capacitance increments obtained over time within the predetermined period (period TP**1**); and the capacitance increments in the corresponding previous scanning processes reaches or exceeds the predetermined value Cb the predetermined number of times or more, transition to the normal mode is performed. In the example as illustrated in FIG. **5**B, the period TP**1** is started from the start point that is the scanning process in which the capacitance increment exceeds Ca. In the fourth modified example of the present embodiment, as an example, a scanning process in which the capacitance increment exceeds a predetermined value is the start point (in FIG. **12**, it is assumed that the capacitance increment exceeds the predetermined value in the first scanning process). However, this is by no means a limitation. The start point of the period TP**1** may be the time the cumulative value of the differences between the increments exceeds the predetermined value or may be the time the capacitance exceeds the predetermined value. Also, in addition to the time the capacitance increment or the cumulative value of the differences between the increments exceeds the predetermined value as described above, the start point of the period TP**1** may be set at predetermined time intervals.
(110) The cumulative value as used in the present disclosure may be a cumulative value of target numerical values (differences between the capacitance increments in the fourth modified example of the embodiment as described above) during the previously set scanning period or at the previously set number of times of scanning processes. Also, the scanning period or the number of scanning processes for cumulation is set to an appropriate period or number of times in accordance with the detection level of the capacitance, the scanning conditions, and the like. For example, the scanning period for which the cumulative value is obtained may be the same as or shorter than the above predetermined period (period TP**1**). Also, the cumulative value obtained through cumulation that is started from a period before the above predetermined period (period TP**1**) may be set. The cumulative value is cleared at the end of the scanning period for cumulation or the number of times of scanning processes for cumulation.
(111) When the electrostatic sensor **110** includes the sensors **111**, **112**, and **113**, the difference from the capacitance increments in the previous scanning process can be obtained as a difference between an average value of the capacitance increments in the previous scanning process and an average value of the capacitance increments in the current scanning process.
(112) The cumulative value is obtained as follows. For example, when four scanning processes are performed, a difference Cα**1** is determined by subtracting the average value of increments ΔC**11**, ΔC**12**, and ΔC**13** obtained in the first scanning process from the average value of increments ΔC**21**, ΔC**22**, and ΔC**23** obtained in the second scanning process.
(113) Similarly, a difference Cα**2** is determined by subtracting the average value of the increments ΔC**21**, ΔC**22**, and ΔC**23** obtained in the second scanning process from the average value of the increments ΔC**31**, ΔC**32**, and ΔC**33** obtained in the third scanning process.
(114) Similarly, a difference Cα**3** is determined by subtracting the average value of the increments ΔC**31**, ΔC**32**, and ΔC**33** obtained in the third scanning process from the average value of the increments ΔC**41**, ΔC**42**, and ΔC**43** obtained in the fourth scanning process.
(115) Then, the cumulative value (total value) of the differences Cα**1**, Cα**2**, and Cα**3** is determined. When this cumulative value is equal to or more than the predetermined value Cb (an example of

the second predetermined value), transition to the normal mode is performed. The predetermined value here is a value indicating how closely the user has approached the door handle **10**.

(116) Instead of the average value of the increments obtained in each of the scanning processes, the maximum value or the total value of the increments of the three sensors obtained in each of the scanning processes may be used. A calculation method of the increments or the representative value of the capacitances in the scanning processes may be appropriately determined in accordance with a system employed.

(117) FIG. **13** is a flowchart illustrating a process executed by the controller **126** in the fourth modified example of the embodiment. The flowchart as illustrated in FIG. **13** is the same as the flowchart as illustrated in FIG. **9** except that the processes of steps S**5** and S**8** are changed. Thus, only the difference from FIG. **9** will be described.

(118) In step S**5**A, the controller **126** determines whether or not the cumulative value of the differences between the average value of the capacitance increment obtained in the previous scanning process and the average value of the capacitance increment obtained in the current scanning process is the predetermined value Cb or more (step S**5**A).

(119) In step S**8**A, when the controller **126** determines in step S**5**A that the cumulative value is not the predetermined value Cb or more (S**5**A: NO), the controller **126** determines whether or not the duration time is the period TP**1** or longer (step S**8**A).

(120) In the flowchart as illustrated in FIG. **13**, the scanning process in step S**1** is repeated. As a result, when the cumulative value of the differences reaches or exceeds the predetermined value Cb three times or more within the period TP**1**, it is determined to be YES in step S**7** and then transition to the normal mode is performed in step S**3**, thereby activating the electrostatic sensor **110**. The number of times based on which it is determined to be YES in step S**7** may be appropriately set. For example, the electrostatic sensor **110** may be activated even when the cumulative value reaches or exceeds the predetermined value Cb only once.

(121) The above-described configuration uses the cumulative value of the differences of the capacitance increments from the capacitance increments obtained in the previous scanning process. However, the same process may be performed using the capacitance increments rather than the differences therebetween. For example, when the cumulative value of the capacitance increments ΔC**1**, ΔC**2**, and ΔC**3** reaches or exceeds the predetermined value Cb the predetermined number of times, transition to the normal mode is performed. At this time, the predetermined value Cb for the cumulative value may be set to increase stepwise.

(122) At this time, only the capacitance increments ΔC**1**, ΔC**2**, and ΔC**3** that are the predetermined value Ca or more may be used, and when the cumulative value thereof reaches or exceeds the predetermined value Cb, transition to the normal mode may be performed.

(123) Alternatively, when the capacitance increment ΔC**1** is the predetermined value Ca or more and a relation of ΔC**1**<ΔC**2**<ΔC**3** is fulfilled, the cumulative value of ΔC**1**, ΔC**2**, and ΔC**3** may be calculated and used for determination.

(124) In this case, when relations of Cα<ΔC**1**, Cb<ΔC**2**, and Cc<ΔC**3**, and a relation of Cα<Cb<Cc are fulfilled, the cumulative value of ΔC**1**, ΔC**2**, and ΔC**3** may be calculated and used for determination.

(125) It is possible to provide an electrostatic input device that is capable of activating the capacitance detector in a state before determination of contact, and suppresses false detection.

(126) Although the electrostatic input device of the illustrative embodiment of the present invention has been described above, the present invention may be any combination of the specifically disclosed embodiments.

(127) The present invention is not limited to the specifically disclosed embodiments, and can be changed and modified in various ways without departing from the claims recited.

Claims

1. An electrostatic input device, comprising: a capacitance detector that is disposed at an operation part to be operated by a living body, and is configured to repeatedly detect capacitance; and a controller that is formed of an integrated circuit, and is configured to transition the capacitance detector to an activated state in response to an increment in the capacitance detected by the capacitance detector reaching or exceeding a first predetermined value a plurality of times within a predetermined period in a case in which the capacitance detector is in a standby state and no contact with the living body is detected, the increment in the capacitance is a difference between a capacitance value detected in a previous detection and a capacitance value detected in a current detection among capacitance values repeatedly detected by the capacitance detector, the difference being calculated by subtracting the capacitance value detected in the previous detection from the detected capacitance value detected in the current detection.
2. The electrostatic input device according to claim 1, wherein the controller includes a counter configured to increment a value in response to the increment in the capacitance reaching or exceeding the first predetermined value, and the controller is configured to transition the capacitance detector to the activated state in response to a count value of the counter reaching a predetermined count value within the predetermined period, the predetermined count value corresponding to the predetermined number of times.
3. The electrostatic input device according to claim 2, wherein in response to the count value of the counter not reaching the predetermined count value within the predetermined period, the controller is configured to clear the count value of the counter in response to the increment in the capacitance reaching or falling below a third predetermined value that is greater than the first predetermined value, or set a second predetermined value that is less than the predetermined count value without clearing the count value in response to the increment in the capacitance reaching or exceeding the third predetermined value.
4. The electrostatic input device according to claim 1, wherein the controller is configured to transition the capacitance detector to the activated state in response to an increase over time in the increment in the capacitance that reaches or exceeds the first predetermined value.
5. The electrostatic input device according to claim 1, wherein the controller is configured to increase the first predetermined value stepwise within the predetermined period.
6. The electrostatic input device according to claim 1, further comprising: the operation part.
7. The electrostatic input device according to claim 1, wherein the controller is configured to set the capacitance detector to be in the standby state without the living body contacting the operation part, and set the capacitance detector to be in the activated state with the living body contacting the operation part.
8. The electrostatic input device according to claim 1, wherein the capacitance detector includes multiple sensors, and the controller is configured to transition the capacitance detector to the activated state using, as the capacitance, a maximum value, an average value, or a total value of multiple capacitances that are detected by the multiple sensors the predetermined number of times or more within the predetermined period.
9. An electrostatic input device, comprising: a capacitance detector that is disposed at an operation part to be operated by a living body, and is configured to repeatedly detect capacitance; and a controller that is formed of an integrated circuit, and is configured to transition the capacitance detector to an activated state in response to a difference between increments in the capacitance consecutively detected by the capacitance detector reaching or exceeding a first predetermined value a plurality of times within a predetermined period in a case in which the capacitance detector is in a standby state and no contact with the living body is detected, the increment in the capacitance is a difference between a capacitance value detected in a previous detection and a

capacitance value detected in a current detection among capacitance values repeatedly detected by the capacitance detector, the difference being calculated by subtracting the capacitance value detected in the previous detection from the detected capacitance value detected in the current detection, the difference between the increments in the capacitance being calculated by subtracting an increment in the capacitance obtained in a previous detection from an increment in the capacitance obtained in a current detection.

10. The electrostatic input device according to claim 9, wherein the controller includes a counter configured to increment a value in response to the difference reaching or exceeding the first predetermined value, and the controller is configured to transition the capacitance detector to the activated state in response to a count value of the counter reaching a predetermined count value within the predetermined period, the predetermined count value corresponding to the predetermined number of times.

11. The electrostatic input device according to claim 9, wherein the controller is configured to transition the capacitance detector to the activated state in response to the difference reaching or exceeding the first predetermined value in conjunction with the difference that reaches or exceeds the first predetermined value increasing over time.

12. The electrostatic input device according to claim 9, wherein the controller is configured to increase the first predetermined value stepwise within the predetermined period.

13. An electrostatic input device, comprising: a capacitance detector that is disposed at an operation part to be operated by a living body, and is configured to detect capacitance; and a controller that is formed of an integrated circuit, and is configured to transition the capacitance detector to an activated state in response to an increment in the capacitance detected by the capacitance detector reaching or exceeding a first predetermined value a predetermined number of times within a predetermined period in a case in which the capacitance detector is in a standby state and no contact with the living body is detected, wherein the controller includes a counter configured to increment a value in response to the increment in the capacitance reaching or exceeding the first predetermined value, and the controller is configured to transition the capacitance detector to the activated state in response to a count value of the counter reaching a predetermined count value within the predetermined period, the predetermined count value corresponding to the predetermined number of times.